



Hadoop Cluster

Architecture & Challenges

Design decisions, node roles & lessons learned from a 5-node HA cluster

Hadoop Version	Hadoop 3.4.2
ZooKeeper	ZooKeeper 3.8.6
Java Runtime	OpenJDK 17
OS	Ubuntu 24.04
Environment	Docker Containers
Cluster Size	5 Nodes (node01 – node05)
HA Mode	HDFS QJM + Automatic YARN Failover

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1. Cluster Architecture

Overview

A 5-node Hadoop HA cluster running in Docker containers with automatic failover for both HDFS and YARN. Built using Hadoop 3.4.2, ZooKeeper 3.8.6, OpenJDK 17, and Ubuntu 24.04.

Node Roles

Node	HDFS	YARN	Coordination
node01	NameNode (Active)	ResourceManager	ZooKeeper, JournalNode, ZKFC
node02	NameNode (Standby)	ResourceManager	ZooKeeper, JournalNode, ZKFC
node03	DataNode	NodeManager	ZooKeeper, JournalNode
node04	DataNode	NodeManager	—
node05	DataNode	NodeManager	—

Why This Distribution?

- ▶ **Master Co-location:** NameNode and ResourceManager are co-located on node01 & node02 because both are master-level services. Separating NameNode from DataNode/NodeManager is essential — you don't want compute tasks competing for resources with the metadata service.
- ▶ **Data Locality:** DataNode and NodeManager are co-located on node03, node04, and node05 for data locality. When YARN schedules a task, it prefers to run it on a node that already has the data blocks — the task reads data locally instead of transferring it over the network, significantly improving performance.
- ▶ **ZooKeeper Quorum:** ZooKeeper runs on node01, node02, node03 because it needs an odd number (3) to form a quorum. With 3 nodes, the cluster tolerates 1 ZK failure.
- ▶ **JournalNodes & QJM:** JournalNodes run on node01, node02, node03 alongside ZooKeeper to keep the coordination layer together. JournalNodes store the shared edit logs between active and standby NameNodes using QJM (Quorum Journal Manager), which requires a majority (2 of 3) to write successfully.
- ▶ **ZKFC Placement:** ZKFC runs only on node01 & node02 — it monitors the local NameNode's health and communicates with ZooKeeper to manage automatic failover. It only exists where NameNodes exist.

2. Challenges Faced & Solutions

Challenge 1: Installing Hadoop on Docker Containers

Hadoop and OpenJDK must be installed manually inside containers after composing the image. The setup process was guided by the Apache Hadoop 3.4.2 official documentation and community resources for Ubuntu 24.04.

Challenge 2: Environment Variables Setup

Initially attempted to mount an env folder like a pre-built image project. With bare Ubuntu containers, everything had to be configured manually. `/etc/bash.bashrc` (system-wide) is more reliable in Docker than per-user `.bashrc` for persistent environment variables.

✓ **Fix:** Use `/etc/bash.bashrc` for system-wide environment variables in Docker containers.

Challenge 3: Missing Configuration Properties

The official Hadoop HA with QJM documentation did not cover all mandatory configuration properties. Critical settings (fencing method, auto-failover, QJM shared edits directory) were missing from the docs. Each property had to be researched individually before being added to the config files.

✓ **Fix:** Cross-reference the official docs with a working reference cluster project to identify all required HA properties.

Challenge 4: `docker-compose down` Destroys Everything

Running `docker-compose down` deletes all containers and everything installed inside them, losing all Hadoop setup.

✓ **Fix:** Use ``docker commit`` to save each container's state as a Docker image. Combine with startup scripts that automatically bring services up on ``docker-compose up``.

Challenge 5: JournalNodes Dying from SIGHUP

Using ``docker exec -it bash -l -c "command"`` killed daemons silently. The login shell (`-l` flag) sends SIGHUP to child processes on exit, terminating long-running daemon processes unexpectedly.

✓ **Fix:** Use ``docker exec bash -c "source ... && command"`` without `-it` or `-l` flags for all daemon commands.

Challenge 6: NameNode Format Requires JournalNodes Running

In HA mode, the NameNode format communicates with JournalNodes through QJM. If JournalNodes aren't running, the format fails with "Connection refused." The strict startup order must be followed: ZK → JN → Format → NN.

✓ **Fix:** Always start ZooKeeper first, then JournalNodes, then format, then NameNodes — in that exact order.

Challenge 7: Docker Blocks `kill -9`

Hadoop's stop scripts use kill -9 as a fallback, but Docker containers block SIGKILL by default, preventing Hadoop from gracefully stopping its own processes.

✓ **Fix:** Add `privileged: true` to each service in docker-compose.yml to allow SIGKILL.